

Format

Reading Time: 10 mins

Working Time: 100 mins

Structure

Section 1: 55 multiple choice

Section 2: 13 short answer

Section 3: Reading comprehension - 6 questions

Areas of Study

Astronomy

- Definitions.
- Distances in space - no light year calculations.
- Star lifecycles.
- Doppler Effect and red shift/blue shift.
- Hubble's observations of red shift.
- Origin of the universe - Big Bang versus Steady State.
- Einstein's equation - $E = mc^2$ - relationship to matter being created in the early universe. No calculations involved.

Genetics

- Definitions.
- Dominant/recessive traits.
- Incomplete/co-dominant traits.
- Punnett squares.
- Pedigrees.
- Mitosis/meiosis.
- Sex inheritance.
- Karyotypes.
- Mutations.
- Reproduction - sexual versus asexual, connection to mitosis/meiosis.